APEXLOGIX SOLUTIONS: SYSTEM ARCHITECTURE DESIGN

1. Architecture Overview

The ApexLogix platform utilizes a highly modular microservices design to scale telemetry ingestion and robot path routing. The layout consists of an API Gateway reverse proxy routing incoming traffic to specialized domain services. PostgreSQL is used for relational transactional storage, while TimescaleDB captures the high-frequency robot telemetry metrics.

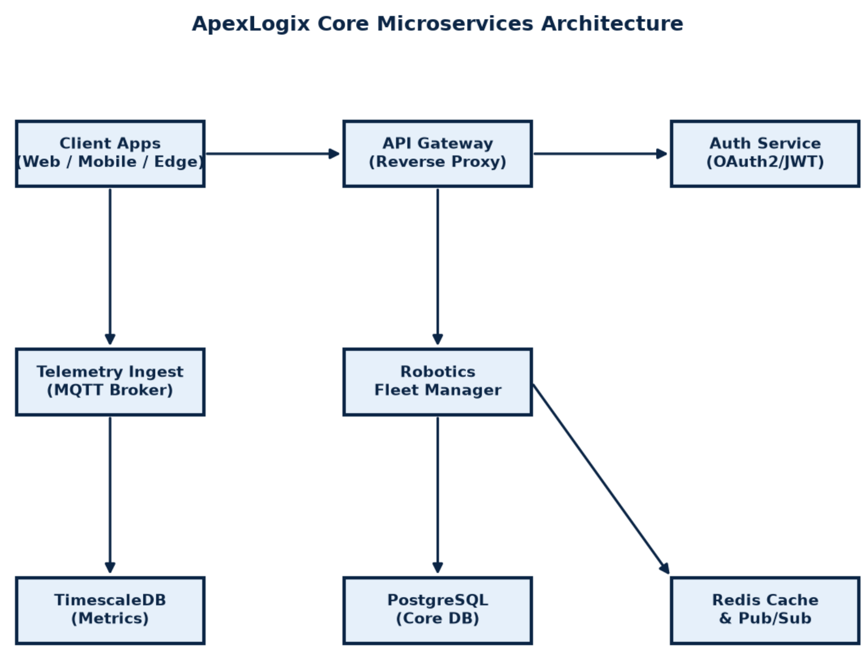

Figure 3.1: Microservices and Storage Layer Layout Diagram.

2. Robot Telemetry Table Schema

Telemetry reports are streamed via MQTT every 500ms from active robots. The metrics are ingested and stored in the relational timeseries table structured as follows:

Column Name	Data Type	Constraint & Description
id	BIGINT	PRIMARY KEY, auto-incrementing unique identifier.
robot_id	VARCHAR(36)	UUID referencing the unique active robot registration.

timestamp	TIMESTAMPTZ	UTC time coordinates of packet transmission.
battery_level	NUMERIC(4,2)	Percentage value indicating battery power (0.00% to 100.00%).
x_coord / y_coord	REAL	Position coordinates relative to the warehouse grid coordinates.
status_code	INTEGER	Status enum value: 0=Idle, 1=Routing, 2=Charging, 3=Fault.
error_log	TEXT	Optional field containing error codes or system traceback dump.